BM3122 | MEDICAL IMAGING

DOCUMENT D

Final revision and prediction guide

A practical study plan based on three independent past papers and the supplied syllabus. High confidence refers to preparation value, not certainty about a future question.

1. Priority-ranked question families

1. MRI spatial encoding and sequence timing [High]

Three years of spatial encoding; diagrams recur in latest two. Evidence: 2023 3(b)ii-iii; 2024 3(b)i-ii; 2025 3(a)ii-iii,3(b)i-iv. Revise: Larmor Hz/rad conversion; full bandwidth; TR*Nphase; GE/multislice diagrams.

2. Ultrasound depth, resolution and frame rate [High]

Numerical family appears in all three papers. Evidence: 2023 4(a)iii-iv; 2024 4(a)iv-vi; 2025 4(a)iii-v. Revise: 2*attenuation*f*d; c/(2d*Nlines); pulse length; focus.

3. Nuclear decay and count fractions [High]

A numerical decay task occurs every year with changing initial conditions. Evidence: 2023 1(c)i-ii; 2024 1(b)iv; 2025 1(b)iv. Revise: A=lambda*N; effective half-life; logarithms; count efficiency assumptions.

4. X-ray attenuation, system and CT reconstruction [High]

Complete 25-mark Q2 block every year; subtopic rotation prevents narrow selection. Evidence: 2023 2(a),2(b); 2024 2(a)iii,2(b); 2025 2(a),2(c). Revise: Tube/grid diagram; exponential attenuation; Fourier-slice diagram; ART residual.

5. Image quality: PSF/LSF [High]

Latest two years dedicate image-quality questions to spatial resolution. Evidence: 2024 1(a)ii-iii; 2025 1(a). Revise: Gaussian half-height algebra; delta and top-hat convolution; MTF.

6. PET/SPECT and radiotracer selection [High]

Two recent papers use explanations alongside decay. Evidence: 2024 1(b)i-iii; 2025 1(b)i-iii. Revise: 511 keV pair/LOR; collimator sensitivity; uptake; half-life.

7. US artifacts and compounding [High]

Direct artifact emphasis in latest two papers. Evidence: 2024 4(a)ii-iii; 2025 4(a)ii. Revise: Interference vs random electronic noise; causal ray sketches.

8. OCT basics and application [High]

Two recent optical blocks; no supplied lecturer OCT slides. Evidence: 2024 4(b)i-ii; 2025 4(b)i-iii. Revise: Source/coupler/reference/sample/detector; path matching; retinal layers.

9. GRE Ernst angle and SE comparison [Medium]

Strong one-year numerical precedent plus recent sequence comparison. Evidence: 2024 3(b)iii-v; 2025 3(b)v-vi. Revise: Spoiled steady-state assumption; fixed TR/TE/noise; T2 vs T2*.

10. DSA, HU calibration, CT generations and QA [Medium]

Each is evidenced, but alternatives rotate. Evidence: 2023 2(a),2(b)ii-iii; 2025 2(b). Revise: Log sign; water/air reference; geometry; four QA checks.

11. ADC and thermal imaging [Medium]

Absent in latest two but explicit syllabus and genuine precedent. Omission alone does not increase priority. Evidence: 2023 1(b),4(b). Revise: LSB, gain/clipping; IR optics to surface temperature; complementary use.

12. Additional syllabus coverage [Low]

Protects against overfitting; scope is explicit, recurrence evidence is limited. Evidence: No direct dedicated question in these three papers. Revise: Basic mechanisms; supplied equations only; no advanced signal-analysis demands.

2. Topic-by-topic preparation checklist

□ Modalities, physical signals, structural/function | High

Energy, ionization, pixel meaning; MRI/US pixels are sequence/processing dependent. Source: S1 3-31. Historical check: 2023 1(a); 2024 1(a)i.

□ Spatial resolution; LSF, FWHM | High

Thin-line phantom, half-height Gaussian width. Source: S1 39-44. Historical check: 2024 1(a)ii-iii; 2025 1(a)i.

□ PSF, convolution and MTF | High

I=O*h; anisotropy; fine features blur; MTF magnitude. Source: S1 45-48. Historical check: 2025 1(a)ii-iii.

□ SNR, averaging, photon statistics | Medium

SNR scales as sqrt(counts) or sqrt(independent repeats). Source: S1 50-53. Historical check: 2024 1(b)ii (PET context).

□ Contrast, CNR, partial volume | Medium

Declare contrast definition; CNR=(SA-SB)/sigma. Source: S1 55-56. Historical check: 2024 2(a)iii (contrast).

□ Artifacts | High for US

Physical mechanism distinct from random noise. Source: S1 58-59. Historical check: 2024 4(a)ii-iii; 2025 4(a)ii.

□ ADC, dynamic range, quantization | Medium

n bits; step Vrange/2^n; gain without clipping. Source: S1 62-66. Historical check: 2023 1(b)i-iii.

□ Sampling, oversampling, filtering | Coverage

Nyquist; anti-alias filter; fourfold oversampling per bit under assumptions. Source: S1 67-74. Historical check: No direct subquestion.

□ X-ray tube, focal spot, heel, spectrum | High

f=F sin(theta); heat/coverage; kVp/mAs; line+continuum spectrum. Source: S2 7-15. Historical check: 2025 2(a)i-iii.

□ Photon interactions | Medium

Photoelectric vs Compton vs Rayleigh; pair threshold. Source: S2 16-26. Historical check: 2024 2(a)ii.

□ Attenuation, mass mu, HVL | High

Exponential law and sum of line integrals; HVL=ln2/mu. Source: S2 27-31. Historical check: 2023 2(a); 2024 2(a)iii; 2025 2(a)iv.

□ System, collimator, grid, CR/DR | High / coverage

Grid contrast/dose trade-off; scintillator vs direct charge conversion. Source: S2 32-42. Historical check: 2025 2(a)i.

□ Contrast agents and DSA | High

Barium vs iodine; log subtraction; matched motion/exposure. Source: S2 43-49. Historical check: 2023 2(a)i-iv; 2024 2(a)i.

□ Mammography, fluoroscopy | Coverage

Small focus, compression; dynamic acquisition and dose. Source: S2 50-53. Historical check: No direct subquestion.

□ Dose and protection | Coverage

Gy vs Sv; tissue weighting; time/distance/shielding. Source: S2 54-59. Historical check: No direct subquestion.

□ X-ray image examples | Coverage

Relate attenuation/contrast to anatomy; examples, not diagnoses to memorize. Source: S2 60-70. Historical check: No direct image-identification subquestion.

□ CT projections, Radon, sinogram | High

-ln(I/I0); disk chord length; projection angle vs detector coordinate. Source: S3 2-13. Historical check: 2023 2(b)i.

□ Central section theorem, Fourier reconstruction | High

Projection FT is radial slice of object 2D FT. Source: S3 14-16. Historical check: 2024 2(b)ii; 2025 2(c).

□ Backprojection, FBP, ART, MLEM | High

Ramp filter; ray residual distribution; iterative forward projection. Source: S3 17-29. Historical check: 2024 2(b)iii; 2025 2(c).

□ HU and calibration | High

Water=0, air approximately -1000; linear calibration. Source: S3 31-32. Historical check: 2025 2(b).

□ CT generations, helical, multislice, dual energy | Medium / coverage

Motion geometry; pitch convention; detector rows; dual source vs dual energy. Source: S3 35-43. Historical check: 2023 2(b)ii.

□ Special reconstruction, detectors | Coverage

Rebinning, interpolation, thin/thick slices; gantry signal path. Source: S3 45-54. Historical check: 2025 2(a)i only shared radiography hardware.

□ CT artifacts, applications, dose | Coverage

Ring/streak/cupping; motion; clinical context; dose trade-offs. Source: S3 56-77. Historical check: 2023 QA adjacent, not direct artifact question.

□ CT QA | Medium

HU accuracy, uniformity, resolution, thickness; confirm local QA teaching. Source: No explicit QA slide; S3 31-32 context. Historical check: 2023 2(b)iii.

□ NMR, populations, Larmor, RF, detection | High

Net population difference, Hz vs rad/s, tip angle, Faraday induction. Source: S4 7-28. Historical check: 2023 3(b)i-iii; 2024 3(b)iii.

□ Relaxation, chemical shift, FID | Medium

T1 recovery; T2 irreversible; T2* includes static dephasing. Source: S4 29-43. Historical check: 2024 intensity/2025 SE mechanism context.

□ Slice selection, phase/frequency encoding | High

Full RF BW=(gamma/2pi)G delta-z; phase retains position information. Source: S4 44-60. Historical check: 2023 3(b)ii-iii; 2024 3(b)i; 2025 3(a)ii-iii,3(b)i.

□ GE timing, acquisition time, multislice | High

TR*Nphase*NEX; RF center changes between slices; gradients overlap correctly. Source: S4 61-74. Historical check: 2024 3(b)i-ii; 2025 3(b)i-iv.

□ Contrast, Ernst angle, SE, 3D, EPI, TSE | High / coverage

GRE steady state; optimum not always 90 degrees; SE refocuses static offsets. Source: S4 75-86. Historical check: 2024 3(b)iv-v; 2025 3(b)v-vi.

□ k-space, FOV, aliasing, SNR | Medium

FOV=1/delta-k; finite kmax limits detail; voxel/SNR/time trade-off. Source: S4 66-71,87-92. Historical check: 2025 scan-time application.

□ MRI hardware, magnet types, coils | Medium

Main B0; three gradients; RF Tx/Rx; homogeneity, stability. Source: S4 93-101. Historical check: 2023 3(a)ii-iii.

□ MR angiography and fMRI | Coverage

Inflow contrast; BOLD haemodynamic proxy; EPI and task/rest correlation. Source: S4 102-108; syllabus p.1. Historical check: No direct subquestion.

□ US physics, impedance, echo modes | High

Z=rho*c; intensity reflection coefficient is pressure coefficient squared. Source: S05 6-19. Historical check: 2023 4(a)ii; 2025 4(a)i.

□ TGC, piezoelectricity, arrays, focusing | Medium

Tx/Rx reciprocity, delays, focal beam width, TGC cannot recover lost echoes. Source: S05 20-25,30-31,40. Historical check: 2024 4(a)iv (arrays).

□ US resolution, penetration, timing | High

Round trip 2d; axial=Nc*c/(2f); frame=c/(2d*Nlines). Source: S05 16-19,26-29. Historical check: 2023 4(a)iii-iv; 2024 4(a)iv-vi; 2025 4(a)iii-v.

□ US compounding and artifacts | High

Coherent speckle; angular/frequency compounding; shadow/enhancement/reverberation. Source: S05 32-38. Historical check: 2024 4(a)ii-iii; 2025 4(a)ii.

□ Doppler | Coverage

Moving blood causes frequency shift; direction dependence; no detailed equation in slide. Source: S05 39; syllabus p.1. Historical check: No direct subquestion.

□ NM activity, half-life, radiotracers | High

A=lambda*N; effective rate sums; activity vs atom count vs detected counts. Source: S06 2-10. Historical check: 2023 1(c); 2024 1(b); 2025 1(b)iv.

□ Gamma camera, collimation, dead time, planar | Coverage

Direction selection, sensitivity penalty; nonparalyzable count correction. Source: S06 11-19. Historical check: No direct equipment/dead-time subquestion.

□ SPECT, correction, SPECT/CT | Medium

Rotating projections; scatter energy window; attenuation correction. Source: S06 20-24. Historical check: 2024 1(b)i-ii.

□ PET, coincidence, PET/CT, detectors, 2D/3D | High

511 keV pair, LOR, true/random/scattered events, functional/anatomical fusion. Source: S06 25-40. Historical check: 2024 1(b)ii-iii; 2025 1(b)i-iii.

□ OCT principle, scans, resolution, applications, QA | High basics; assignment-only depth Medium

Low-coherence interferometer; optical backscatter; A/B scans; shallow depth. Source: Syllabus p.1; A2 PDF 3-8. Historical check: 2024 4(b); 2025 4(b).

□ Thermal principle, applications, QA | Medium

Emitted IR to detector/ADC/calibrated surface temperature; environment and emissivity. Source: Syllabus p.1; A1 PDF 3-9. Historical check: 2023 4(b).

Completion standard: explain the mechanism without notes, write the governing equations with units, draw the key schematic, and solve one changed-data problem. “Coverage” items need a minimum conceptual pass even when their exact historical frequency is zero. No verified lecturer exclusions were found.

3. Mathematical preparation

Family	Equation / method and source
Gaussian width	Set L/Lmax=1/2; take ln; get two roots; subtract them: FWHM=2√(2 ln2)σ. S1 44.
PSF / MTF	I=O*h for linear shift-invariant systems. Unit-area h preserves integrated signal; normalized MTF is the magnitude of the optical transfer response. Distinguish directional PSFs. S1 45-48.
CNR and averaging	CNR=(SA-SB)/σN; mixture signal=sum of volume fractions times signals. Independent averages improve SNR as √N; coherent speckle needs decorrelation. S1 52-56.
ADC / sampling	Step=Vspan/2^n for specified equal-width bins; ideal rounding error≤step/2. fs≥2fmax for a baseband signal, with practical anti-alias margin. Fourfold oversampling per ideal effective bit requires noise assumptions. S1 62-72.
X-ray attenuation and HVL	I=I0 exp(-Σμixi); μ=ln2/HVL; p=-ln(I/I0). Always match cm to cm^-1 or m to m^-1. S2 27-31; S3 4.
DSA algebra	I2-I1 retains exp(-μtxt); lnI2-lnI1=-(μc-μb)xv when incident exposure, geometry and static tissue are unchanged. S2 48-49; 2023 Q2(a).
CT projection / theorem	Uniform disk p(s)=2μ0√(R²-s²) for |s|≤R. 1D FT of projection is radial slice of 2D object FT. Explain chord geometry, not just memorized formula. S3 4,11-16.
ART	Compute predicted ray sum, residual=measured-predicted, then divide by pixel count for stated unit weights. Track every updated image. Sparse projections can fit many images. S3 26-28.
HU / calibration	HU=1000(μ-μwater)/μwater; μ=μwater(1+HU/1000). Solve two simultaneous reference equations for a,b in h=a hm+b. S3 31-32.
Focal spot / coverage	f=F sinθ; coverage=2D tanθ. Keep angular units consistent. Smaller projected focus trades against field coverage and/or loading. S2 9-13.
Dose / pitch	D=energy/mass; H=wR D; E=ΣwT HT. Gy is energy/mass; Sv is a weighted measure. Helical pitch convention must specify total beam width for MDCT. S2 54-59; S3 39-41.
Radioactive decay	A=λN; λ=ln2/Thalf. Stock back-extrapolation uses A0=A(t) exp(λt). Activity ratio evolves as (λ1N10/λ2N20) exp[-(λ1-λ2)t]. S06 6-7.
Effective half-life / dead time	λeff=λphys+λbio. For non-paralyzable counting N=n/(1-nτ), nτ<1. Count fractions require efficiency assumptions; atom equality is not activity equality. S06 8,14-17.
Larmor / slice bandwidth	f0=γB0/(2π); full Δf=(γ/2π)|G|Δz. Central endpoints=f0±Δf/2; displaced slice centre uses B0+Gzcentre. 1 mT/cm=0.1 T/m. S4 15,48-54.
Polarization / RF	Population imbalance fraction=γhB0/(4πkT); Kelvin required; multiply by 100 for %. Flip angle depends on integrated B1 pulse. S4 12-22.
GE optimum	E1=exp(-TR/T1); GE signal proportional to (1-E1)sinα/(1-E1cosα) times exp(-TE/T2*). Differentiate to obtain cosαE=E1. At fixed noise, fractional SNR loss is 1-Sother/Sopt. S4 78.
SE / acquisition / k-space	SE envelope uses exp(-TE/T2). Tscan=TR Nphase NEX for conventional 2D. Interleaved capacity=floor(TR/full sequence slot), not universally TR/TE. FOV=1/Δk. S4 61-92.
US propagation and reflection	λ=c/f; Z=ρc. Pressure reflection=(Z2-Z1)/(Z2+Z1), intensity reflection is its square. Round-trip dB loss=2αfd. Additional intensity reflectance R contributes -10 log10 R loss. S05 11-16.
US timing and resolution	tline=2d/c; FR≤c/(2d Nlines Nfirings). Max integer lines must be rounded down. Axial=Nc c/(2f); lateral at focus≈(c/f)(F/L). S05 6-7,17,25-29.
OCT / thermal extension	Gaussian OCT δzair=(2 ln2/π)λ0²/Δλ; tissue depth divide by n. Thermal total-emission ratio=(T2/T1)^4 at equal emissivity; Kelvin required. A2 PDF 5; A1 PDF 4. Assignment-level numerical depth, not past-paper recurrence.
Secondary X-ray equations	Compton shift Δλ=h(1-cosθ)/(mec); mass attenuation coefficient μ/ρ; grid ratio=h/d and grid frequency=1/(d+t). Understand variables and units. No direct numerical precedent for these in the supplied three papers. S2 22,30,36.
Secondary MRI equations	Flip angle α=γ∫B1(t)dt in radians; relative phase φ=γ y∫Gy(t)dt. kx=(γ/2π)∫Gx(t)dt, with k in cycles per length. M0 scales with the net population imbalance; know magnetization versus received-voltage dependence. S4 17,22,57,66-67.

Family	Equation / method and source
Secondary US equations	$c=1/\sqrt{(\kappa\rho)}$, $Z=\rho c=\sqrt{(\rho/\kappa)}$, with κ compressibility. TGC increases gain with depth to offset average attenuation but cannot recreate an absent echo. At a normal interface distinguish pressure transmission from energy transmission. S05 11-16,20.
More syllabus-only practice	Sketch a sinogram of an off-centre point; explain the ramp filter and apodization noise/detail trade-off; interpret the MLEM forward/backprojection update rather than memorize unexplained symbols. Review source-provided equations before assigning numerical practice. S3 8-9,21-29.

4. Diagrams to practise from memory

Diagram	Required labels / practice location
Basic GE sequence	RF with Gslice; slice rephase; variable Gphase; read prephaser/readout; ADC; TE and TR. Models 1 Q3(c), 3 Q3.
Multislice timing	Separate RF centre frequencies in unused per-slice recovery time; retain each slice TR; separate k-space. Models 1 Q3(d), 3 Q3(e).
Spin echo	90° then 180° at TE/2 then echo at TE; distinguish T2 envelope from T2*. Model 1 Q3(f).
PSF convolution and Gaussian LSF	Point to spread profile; rounded step; fine features merge; two half-height roots. Model 1 Q1; Model 3 Q1.
Radiography and CT	Tube→collimator→patient→grid→detector; tube/anode line focus; rotate-rotate vs rotate-stationary; Fourier radial slice. Models 1 Q2, 2 Q2, 3 Q2.
PET and gamma camera	511 keV pair and LOR; window and true/random/scattered events; gamma→collimator→scintillator→photodetectors→position/energy. Models 1 Q1, 4 Q1.
Ultrasound	Pulse/echo time axis; array delays and focus; TGC; shadow/enhancement/reverberation causal paths. Models 1 Q4, 4 Q4.
OCT	Broadband source, splitter, sample arm, reference mirror, recombination detector; optical path matching; A-scans to B-scan. Models 1 Q4, 4 Q4.
Thermal camera	Skin IR→IR optics→detector array→readout/ADC→calibrated temperature display; mark environmental inputs. Model 2 Q4.
Coverage MRI diagrams	Mz recovery/Mxy decay; k-space lines; task/rest with delayed BOLD response. Model 4 Q3.

5. Common examination mistakes

Missing factor two

US echoes travel out and back. RF full bandwidth spans both sides of a central slab. Distinguish full width from half-width.

Wrong scale

mT/cm to T/m is a factor 0.1 per stated numerical value; γ in rad/s/T must be divided by 2π for Hz. Never insert °C in polarization or T^4 laws.

Decay initial conditions

Equal atom numbers imply activity ratio λ1/λ2. Count fractions compare one activity against the sum. Physical stock decay differs from in-body effective loss.

Misread signal equations

GRE steady-state denominator matters. A 90° pulse is not always optimum. Spin echo refocuses static offsets but does not remove intrinsic T2 decay.

Scan time counting

Phase steps repeat the excitation; readout samples occur within a readout. One depth image pixel is not one extra transmit pulse. Multiple slices can interleave rather than multiply scan time.

Confusing noise and artifacts

Identical-view speckle is coherent; independent-average scaling cannot simply be applied. A smooth image can hide reduced detail or temporal information.

Undefined contrast

Declare the requested absolute, Weber or Michelson convention. The model questions specify their numerical definition.

Overclaiming reconstruction

A few exact ray sums do not guarantee a unique object. ART intermediate values depend on update order and weighting.

Uncritical copied answers

Historical formula sheets and handwritten answers contain errors; refer to Document A discrepancy notes and the corrected model sheet.

Unjustified clinical claims

Thermal maps are surface-temperature information, and BOLD is a vascular proxy. Explain the physical measurement and its limits; do not replace it with disease-specific certainty.

6. Model-paper coverage matrix

Topic	Paper 1	Paper 2	Paper 3	Paper 4
Image quality / acquisition	Q1(a-c): PSF, FWHM	Q1(a-c): modalities, ADC	Q1(a-c): anisotropy, CNR, averages	Q1(a-c): ADC, Nyquist, filters
Nuclear medicine	Q1(d-f): tracer, PET, effective decay	Q1(d-f): stock, count fractions, PET/SPECT	Q1(d-f): effective decay, dead time, false coincidences	Q1(d-f): gamma camera, SPECT correction, LOR
X-ray	Q2(a-c): hardware, focus, HVL	Q2(a-d): DSA	Q2(a-b): focus/coverage, grid exposure	Q2(a-d): spectrum, interactions, CR/DR, dose
CT	Q2(d-f): advantages, theorem, ART	Q2(e-g): disk, geometry, QA	Q2(c-e): HU, ART ambiguity, iterative limits	Q2(e-f): helical pitch, artifacts

Topic	Paper 1	Paper 2	Paper 3	Paper 4
MRI	Q3: encoding, GE, multislice, Ernst, SE	Q3: hardware, polarization, bandwidth, timing, QA	Q3: Ernst derivation, contrast, interleaving, SNR	Q3: relaxation, k-space, weighting, aliasing, fMRI, TOF
Ultrasound	Q4(a-e): coupling, speckle, depth, frames, artifacts	Q4(a-d): strengths, interactions, frequency, timing	Q4(a-d): reflection budget, resolution, focusing, compounding	Q4(a-d): PZT, arrays, Doppler, acoustic limits
OCT	Q4(f): interferometer and application	-	Q4(e): axial-resolution extension	Q4(e): time/Fourier domain, QA
Thermal	-	Q4(e-g): principle, applications, QA	-	Q4(f): T^4 ratio and controls

The set broadens coverage but cannot test every slide detail in 400 marks. Remaining self-check prompts: explain mammography compression and focal-spot choice (S2 50-51), fluoroscopy vs static imaging (S2 52-53), fan-beam rebinning/helical interpolation (S3 45-48), MLEM purpose versus ART (S3 29), CT clinical application/dose choice (S3 61-77), 3D MRI/TSE vs EPI (S4 82-86), 1.5D/2D US arrays and TGC (S05 20,30-31), and PET detector/2D-3D sensitivity trade-offs (S06 35-40). Review these even if a timed paper does not ask them explicitly.

7. A realistic revision sequence

Session	Suggested duration	Work and exit criterion
1. Build the four blocks	2-3 h	Use Document A; make one-page maps for Q1 intro/NM, Q2 X-ray/CT, Q3 MRI, Q4 US/optical. Define every signal and unit.
2. Core numerical practice	3 h	Decay ratios/effective half-life; attenuation/HVL/HU; MRI bandwidth/scan time/Ernst; US loss/timing. Redo with changed values and check units.
3. Diagrams and derivations	2 h	Draw GE, multislice, SE, OCT, gamma/PET and X-ray chains from memory. Derive FWHM, DSA log difference and disk chord projection.
4. Minimum full-syllabus pass	2-3 h	Use checklist for fMRI, Doppler, dose, ADC, detectors, QA, MTF, thermal/OCT. Avoid spending all available time on only MRI or repeated numericals.
5. Timed Paper 1	2 h + 1 h review	Attempt closed-book with its data sheet. Mark by proposed allocations; log concept/unit/timing errors. Re-solve missed parts without the answer.
6. Timed Paper 2	2 h + 1 h review	Test rotation resilience: DSA/hardware/thermal. Repair weak mechanisms before proceeding.
7. Paper 3 application practice	2 h + 1-2 h review	Use timing as a diagnostic, since this is intentionally harder. Explain why each numerical answer makes engineering sense.
8. Paper 4 and final repair	2 h + 1 h review	Expose coverage gaps. Revisit the omitted-details prompts and solve previously missed questions from clean sheets.

These sessions require roughly 23-28 focused hours including review and breaks; adapt to your remaining time. If only one day remains, spend about 2 h on equations/units, 1 h on diagrams, 2 h on one timed best-fit paper, 1 h on correction, and 1-2 h on checklist gaps. Use 25-30 minute blocks per modality and short breaks; protect sleep rather than adding a second low-quality full paper.

8. Quality-control record and evidence limits

All 88 historical subquestions reconcile to 100 marks per year. Each model has four 25-mark questions; every proposed rubric balances. Python independently checked 75 numerical results and the ART constraints. PDFs were rendered and visually reviewed. Dedicated OCT/thermal lecture decks and official 2026 instructions/exclusions were unavailable; assignment-supported content is labeled. Predictions are preparation priorities, not guarantees.